

Rahul Rawat

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Dr. Ing. h.c. F. Porsche AG

Hiring Team

Application: Master Thesis Plausibilization of ADAS Front Camera Impairments Using Machine Learning, Kennziffer J000021231

Dear Hiring Team,

I am applying for the Master Thesis on Plausibilization of ADAS Front Camera Impairments Using Machine Learning at Porsche AG in Weissach, Kennziffer J000021231. The brief on inferring impairment hypotheses from contextual, environmental, and vehicle based signals alone, clustering recurring patterns like fogging, icing, or low sun, and developing an interpretable model that assesses plausibility with transparent reasoning maps directly to the interpretable machine learning work I have been shipping over the past year.

In CreditIQ I built an interpretable model that lifted the Disparate Impact ratio from 0.79 to 0.88 against the EU AI Act 80 percent fairness threshold, diagnosed a hidden intersectional bias through SHAP driven subgroup analysis, and cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent. Every decision was documented as a deliberate and regulator defensible trade off with a plain language LLM generated explanation, which is exactly the transparent reasoning your brief calls for.

At eRay GmbH I built a recursive time series pipeline forecasting four water quality indicators for a German lake, benchmarking six models head to head including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and enforcing strict anti leakage rules across the pipeline. In my Economic Impact Analysis of Global Climate Events I ran Random Forest models on tabular event data to translate context signals like duration and severity into calibrated business relevant risk assessments. That is the tabular, context driven, honest evaluation mindset your fleet impairment analysis needs.

I am fluent in English and comfortable in German at B1 in progress, in depth in Python especially pandas, confident with Microsoft Office and Git, and familiar with common statistical frameworks. I hold the NVIDIA Building LLM Applications With Prompt Engineering, AWS Academy Cloud Foundations, and Google Data Analytics Foundations certificates. I would be glad to shape the exact scope with my professor at SRH Heidelberg and start on the schedule that fits your team.

Kind regards,

Rahul Rawat